

Engineering Plasmonic Colloidal Meta-Molecules for Tunable Photonic Supercrystals

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Ordered arrays of metal nanoparticles offer new opportunities to engineer light–matter interactions through the hybridization of Rayleigh anomalies and localized surface plasmons. The generated surface lattice resonances exhibit much higher quality factors compared to those observed in isolated metal nanostructures. Template-induced colloidal self-assembly has already shown a great potential for the scalable fabrication of 2D plasmonic meta-molecule arrays, but the experimental challenge of controlling optical losses within the repeating units has so far prevented this approach to compete with more standard fabrication methods in the production of high-quality factor resonances. In this manuscript, the optical properties of plasmonic arrays are investigated by varying the lattice parameter (between 200 and 600 nm) as well as the diameter of the gold colloidal building-blocks (between 11 ± 1 and 98 ± 6 nm). It is systematically studied how the internal architecture of the repeating gold-nanoparticle meta-molecules influences the optical response of the plasmonic supercrystals. Combining both experimental measurements and simulations, it is demonstrated how, reducing the size of the gold nanoparticles it is possible to switch from strong near-field plasmonic architectures to high-quality factors (>60) for lattice plasmon resonances located in the visible spectral range.

control of nanoparticle size, shape, and composition enables the engineering of localized plasmon resonances (LSPRs), going from the ultraviolet to the far infrared.^[5–7]

An even higher level of tunability of the optical response of metal nanoparticles can be reached controlling the spatial organization of these building blocks, rationally modifying the coupling between their LSPRs.^[8] Coupling nanoparticles in close proximity to one another (i.e., near-field coupling)^[9] generates extremely intense electric fields located between the interacting particles.^[10] The engineering of these hotspots can be achieved by modifying the structure of the nanoparticle clusters (or meta-molecules), and it has led to the development of many surface-enhanced optical detection techniques, such as surface-enhanced infrared absorption,^[11] surface enhanced Raman spectroscopy,^[12,13] or surface-enhanced photoluminescence.^[14,15]

More recently, surface lattice resonances (SLRs) have emerged as a new pathway toward future plasmonic applications.^[16] These lattice resonances are generated when plasmonic nanoparticles or meta-molecules are organized into periodic arrays, enabling the far-field coupling between two physically separated plasmonic objects through the array's diffraction modes.^[17] These collective oscillations show interesting characteristics that are not present in LSPRs, including large area delocalization (hundreds of microns), longer lifetimes, and narrow resonance widths (i.e., high-quality factors).^[16,18,19] Sharp SLRs have been employed for the improvements of the efficiency of various plasmonic detection devices,^[13,20] as well as for the fabrication of antireflection coatings,^[21] more efficient energy harvesting systems,^[22] and for the promotion of nonlinear optical phenomena such as lasing or secondary harmonic generation.^[23–25]

Typically, metal nanoparticle arrays sustaining SLRs are prepared by conventional lithographic processes (e.g., e-beam lithography or photolithography), which are afflicted by elevated fabrication time and costs, and yield polycrystalline structures with higher optical losses. Plasmonic nanoparticles prepared via colloidal synthesis are instead capable of higher optical performances, but their organization with nanoscale precision in large-scale devices is not trivial.^[26] To tackle this issue, several self-assembly approaches have been developed over the years, achieving large-scale organization,^[27,28] high replication

1. Introduction

In the last years, the field of plasmonics witnessed major advances in the fabrication and characterization of more complex architectures, achieving a new level of control over light–matter interactions at the nanoscale.^[1] From a synthetic perspective, the field has evolved from relatively simple spherical metallic colloids to various anisotropic shapes,^[2] heterometallic particles,^[3] and chiral structures.^[4] In return, synthetic

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fidelity, and, in some cases, even single nanoparticle manipulation.^[29] For example, ordered arrays of colloids can be prepared on prepatterned substrates presenting the correct topography using either dip or spin coating setups. Another alternative is represented by capillary-assisted particle assembly^[30] that takes advantage of the capillary forces generated at the contact line with a receding meniscus.^[31] Although these are all viable approaches for the precise organization of colloidal nanoparticles, template-assisted self-assembly emerged as one of the most versatile, especially when combined with nanoimprint lithography, enabling compatibility with roll-to-roll fabrication, and adaptability to different colloids, solvents, and surfaces.^[32–34] The use of polymer substrates also enables the mechanical deformation of the prepared arrays.^[35,36] The most recent developments in the fabrication of this type of structures are reported in Table S1 of the Supporting Information. The highest quality factor reported to date for colloidal plasmonic arrays measured under normal incidence is 48.^[30]

In 2018, we optimized a template-assisted self-assembly methodology for the fabrication of small clusters of gold nanoparticles organized into large-scale arrays.^[13] Exploiting the coupling with the array's diffraction modes, we were capable of accessing array dimensions ranging between 400 and 1000 nm, enabling the wide-spectral tunability of the resulting plasmonic properties.

Herein, we provide full optical characterization of the plasmonic supercrystals fabricated via template-assisted self-assembly of gold colloids. We investigate the role of the meta-molecule and the lattice characteristics in the overall optical response of the plasmonic lattice with particular emphasis on the angular response of the illumination. Moreover, we show the major role played by the meta-molecule architecture in the quality of the produced SLRs, and we demonstrated how this can be engineered changing the size of the assembled colloid, which is easily implemented using template colloidal self-assembly. In particular, assembling gold colloids with a diameter below 25 nm we achieve homogenous arrays of meta-molecules exhibiting resonances with Q factors up to 66, a

major improvement over what previously demonstrated.^[13,31] Our results underpin colloidal self-assembly not only as a versatile and scalable technique, but also as a fabrication approach that enables a new level of control over the plasmonic properties of 2D meta-molecule plasmonic arrays.

2. Results and Discussion

The plasmonic supercrystals presented herein are fabricated via the template-induced self-assembly of gold nanospheres using topographically patterned polydimethylsiloxane (PDMS) molds.^[13] In this work, we employed PDMS molds featuring hole arrays with periodicities between 200 and 600 nm (Figure S1, Supporting Information). The process is summarized in **Figure 1A**: I) a 1 μ L drop of a colloidal suspension of gold nanoparticles ($[\text{Au}^0] = 50 \times 10^{-3} \text{ M}$) coated with thiolated polyethyleneglycol (PEG) is placed over a cleaned glass coverslip; II) immediately after drop-casting, a patterned PDMS stamp is placed above the drop, confining the nanoparticles inside the mold's cavities; III) the solvent is allowed to dry overnight; IV) finally, the PDMS stamp is removed, yielding square arrays of hexagonally packed gold nanosphere clusters (referred to as *meta-molecules* in the rest of the manuscript), forming plasmonic supercrystals with periodicities dictated by the geometry of the stamp, and covering an area of 0.5–1 cm^2 (Figure 1B). It is important to note that the diffractive component of the SLRs is strongly influenced by the refractive index of the propagating media.^[37] The presence of two different refractive indexes for the substrate (glass, $n \approx 1.52$) and superstrate (air, $n \approx 1$) implies a loss of coherence of the SLR, which is translated into a broadening of the peak. For this reason, in the last step of the preparation, all samples are coated with a thin film of PDMS ($n \approx 1.45$), closely matching the refractive index of the glass substrate. Figure 1C summarizes the parameters that can be tuned using our fabrication process: the lattice parameter (Λ), the diameter of the holes engraved in the PDMS stamp (Φ), and the nanoparticle diameter (d).

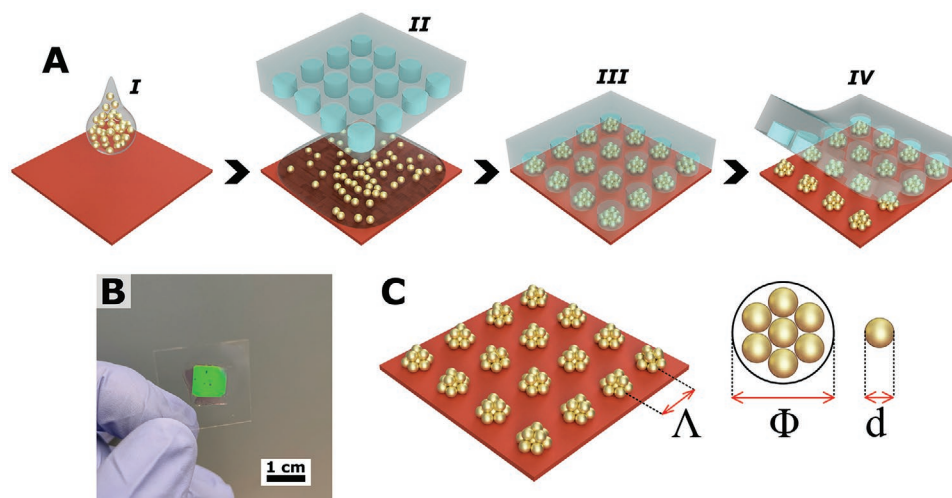


Figure 1. Plasmonic supercrystal fabrication. A) Schematic illustration of the fabrication process of 2D plasmonic meta-molecule arrays. B) Picture of a typical large-scale substrate after preparation. C) Identification of the different parameters characterizing a sample: the lattice parameter (Λ), the PDMS stamp cavity dimension (Φ), and the nanoparticle diameter (d).

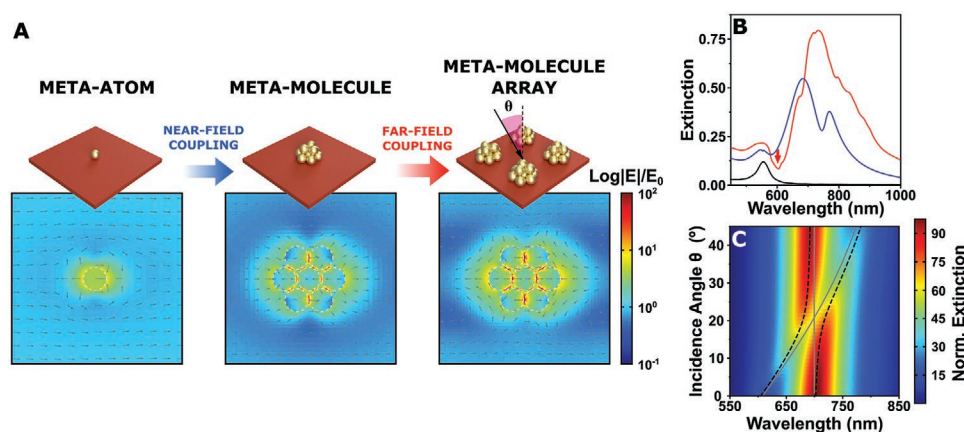


Figure 2. Deconvolution of the optical response of a plasmonic supercrystal made of meta-molecules. A) Schematic (top row) illustrating the coupling mechanisms contributing to the optical properties of meta-molecule-based plasmonic supercrystals. Electric field distribution (bottom row) relative to the extinction maxima of a single nanoparticle (left panel), a single meta-molecule (central panel), and a single meta-molecule as part of an infinite 2D array (right panel). B) Simulated extinction spectra of a single nanoparticle (black solid line), a single meta-molecule (blue solid line), and a meta-molecule 2D array (red solid line, $d = 50$ nm, $\Lambda = 400$ nm, $\Phi = 230$ nm). The red arrow indicates the position of Wood's anomaly. C) Simulated contour plot showing the coupling between a Rayleigh-Wood's anomaly and a localized surface plasmon resonance as a function of the illumination angle θ (more details can be found in the Supporting Information).

Further information relative to the stamps and colloids used in this work can be found in the Supporting Information (Figures S1 and S2, Supporting Information), while the details about colloid and sample preparation are collected in the Experimental Section.

In the case of isolated nanoparticles, the minimum width of the LSPR is limited by radiative damping and the effect of retardation within the particle (depolarization). Optical losses can be overcome disposing the metal nanoparticles in a periodic array, where losses are compensated by the in-phase response of the scattered fields of each nanoparticle upon far-field illumination, translating in significantly narrower SLRs. When the unit cell of the plasmonic lattice is composed of plasmonic meta-molecules instead of single nanoparticles, the optical response is the result of the interplay between the resonances sustained by the meta-molecule and the contribution from the lattice (Figure 2). The isolated meta-molecule sustains near-field coupling between adjacent nanoparticles resulting in coupled LSPRs characterized by high extinction cross-sections (Figure 2B). Moreover, the meta-molecule shows an electric field distribution characterized by the formation of hot-spots that strongly depends on the overall shape and internal organization of the colloidal building-blocks (Figure 2A). The near-field coupling effects and consequently the enhanced electric field regions represent a major advantage of the meta-molecule-based plasmonic supercrystal over a conventional single nanoparticles array, which can be exploited for the amplification of weak optical signals, photothermal effects, or meta-material engineering and, have been responsible for great advances in plasmonic mediated sensing.^[38] Similarly to the case of single particle arrays, these meta-molecule LSPRs interact over long distance, generating delocalized SLRs (far-field coupling, Figure 2A).^[39] In the optical, response of a grating is possible to identify spectral features indicating the offset of diffraction. These are the Rayleigh-Wood's anomalies of the lattice that mark the transition of diffracted modes from evanescent

to propagating.^[40] The characteristic sharp spectral signature of the Wood's anomalies generates Fano-like spectral profile after hybridizing with the plasmonic meta-molecule resonance when they spectrally overlap (red arrow in Figure 2B). Moreover, the far-field coupling between meta-molecules composing the array has important consequences on the electric field distribution of the SLRs, with the regions of highest electric field enhancement extending further away from the meta-atom surface (Figure 2A),^[13] ultimately reducing optical losses. Finally, the presence of a diffractive contribution will also introduce a strong dependency on the illumination angle θ (Figure 2C). It is important to remember that the stronger diffracted wave is the one relative to the 0th order of diffraction, which presents an unmodified tangential component, and its outcome angle matches the pure Snell refraction. This is the strongest order, and will be the one discussed in the rest of the manuscript.

It is worth remembering another effect introduced by the 2D nature of the prepared diffraction patterns, namely, the variation of the Wood's anomalies angular dependency as a function of the azimuthal angle ϕ (Figure S3, Supporting Information). This effect has to be taken into account in order to correctly identify the Wood's anomalies in experimental extinction spectra. All the diffraction lines that are shown in the manuscript (white lines in the contour plots) were calculated by taking into consideration the polarization of the incident field respect to the azimuthal angle of the sample that was used for the measurements.

The most straightforward way to engineer the optical properties of the photonic plasmonic supercrystal is through variations of the lattice parameter. In our system, this is done by the choice of the PDMS stamp used in the preparation.^[13] The influence of the lattice parameter (Λ) and meta-molecule diameter (Φ) is demonstrated in Figure 3, where we show the angular response of 2D meta-molecule arrays prepared using gold nanoparticles 54 ± 4 nm in diameter (d) and square lattices with varying pitches (Λ) ranging between 200 and 600 nm

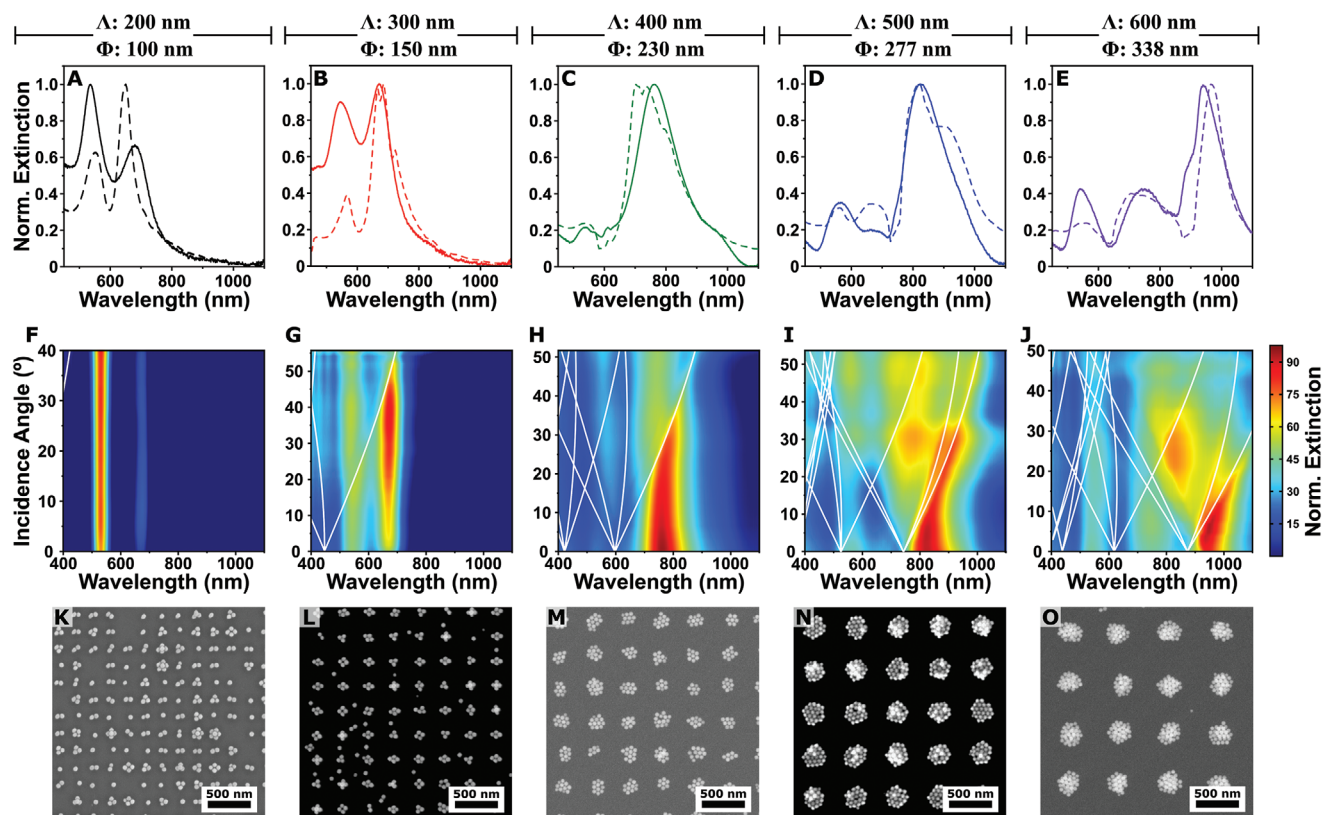


Figure 3. Influence of the lattice parameter and meta-molecule diameter. A–E) Normalized extinction spectra (solid lines) and simulated extinction profiles (dashed lines) at normal illumination incidence, F–J) normalized extinction contour plots as a function of the illumination angle, and K–O) scanning electron microscopy (SEM) images of plasmonic meta-molecule arrays prepared with different lattice parameters: A,F,K) Λ : 200 nm, B,G,L) Λ : 300 nm, C,H,M) Λ : 400 nm, D,I,N) Λ : 500 nm, and E,J,O) Λ : 600 nm. The contour plots have been normalized to the extinction maxima. The nanoparticle dimension is 54 ± 4 nm for all the samples. The white solid lines are analytical calculations of the Wood's anomalies angular dependence. Additional SEM images at different magnifications can be found in Figure S5 in the Supporting Information.

and diameters (Φ) going from 100 until 338 nm (see Table S2, Supporting Information). There are two main contributions to the extinction spectra: a broad angle-independent meta-molecule resonance and the Rayleigh–Woods angle-dependent modes.^[41] At the LSPR frequency, the extinction becomes maximum because the imaginary part of the polarizability of the metal nanoparticle is maximum.^[42] Typically, the real part of this polarizability is positive (negative) for lower (higher) energies than LSPR, causing constructive (or destructive) Fano-like interference, and an extinction peak (or a dip).^[43] Thus, the better figures of merit are expected when the anomalies lie near the meta-molecule LSPR but red-shifted from its central wavelength. The proximity of the Rayleigh–Woods anomalies to the SLR can be selected via the square-grid pitch. This lattice parameter, Λ , is proportional to the diffraction edge wavelengths, and consequently an increase in Λ results in a red-shift of the SLR, which can be observed in the extinction spectra measured under normal incidence (Figure 3A–E). Among this series, we pass from the sub-diffractive case of $\Lambda = 200$ nm to a $\Lambda = 600$ nm where even higher orders of diffraction are notable. In respect to the meta-molecule resonance, a broadening is observed (Figure 3C–E) with increasing Φ , going from 100 or 150 nm (where SEM inspection shows mainly dimers, trimers, and tetramers, Figure 3K,L) up until

227 and 338 nm (where we observed larger meta-molecules, Figure 3N,O).^[44]

The number of nanoparticles within each meta-molecule can be estimated with a simple geometrical model, using only d and Φ as inputs (see Supporting information, Table S3). It is important to underline here that for each combination of d and Φ , one can generate several meta-molecule configurations. This effect, further studied in Figure S11 in the Supporting Information, is confirmed by SEM analysis, and it can definitely contribute to broaden the resonances. However, simulating a 54 nm Au colloid meta-molecule-based plasmonic supercrystal constituted by a single type of meta-molecule confirms the intrinsic broad nature of the associated SLRs, and its red-shift with increasing lattice parameter (Figure S4, Supporting Information).

The interplay between the meta-molecule LSPRs and the Rayleigh–Wood's anomalies can be clearly appreciated when looking at the angular dispersion plot, where the Rayleigh–Wood's anomalies shift and split as a function of the illumination angle (Figure 3F–J). All the expected diffraction lines and their angular dependence can be clearly identified for all lattice parameters. Using the contour plots is possible to recognize the hybridization of the meta-molecule plasmonics with the various diffraction lines. For a Λ of 200 nm, the optical response is practically invariant with the illumination angle, as

the diffraction does not interact with the meta-molecule resonance (Figure 3F). In the case of a Λ of 300 nm, the diffraction clearly couples with the localized plasmonic response when the illumination angle reaches 15° (Figure 3G). The same type of interaction can be identified for larger Λ , with a more complex diffraction pattern.

One of the major benefits of colloidal self-assembly is the possibility to engineer the optical response of the samples by modifying the colloidal building blocks used to fabricate the photonic array. This is particularly appealing, as it represents a parameter that is completely independent from the PDMS stamp. An alternative approach would be to vary Φ while keeping Λ constant. This is certainly valid, but requires additional lithography for the preparation of the masters. For this reason, we did not explore it experimentally, but important theoretical considerations can be found in the Supporting Information (Figure S6). The effect of modifying the meta-molecule architecture by tuning the diameter of the nanoparticle used for the self-assembly is demonstrated in Figure 4. Specifically, we prepared four additional gold nanoparticle colloids, namely: 11 ± 1 , 22 ± 3 , 35 ± 3 , and 98 ± 6 nm in diameter (Figure S2, Supporting Information). The lattice parameter was maintained constant at 400 nm (the case of 500 and 600 nm can be found in the Supporting Information, Figures S8 and S9). Noticeably, by increasing the nanoparticle diameter to 98 nm, we registered a sharpening of the SLR and an additional broad band

around 900 nm (Figure 4E,J,O). In this case, SEM inspection reveals the formation of plasmonic meta-molecules constituted by few-to-single nanoparticles (effect of the dimers and trimers is discussed in Figure S10, Supporting Information). Interestingly, reducing the size of the colloidal had a similar narrowing effect over the SLRs, as can be observed for colloids of 22 and 11 nm (Figure 4A,F,K,B,G,L, respectively).

The observed broadening and narrowing of the SLRs can be rationalized by a combination of two phenomena. On the one hand the coupling between Wood's anomalies and plasmonic resonances generates characteristic Fano-like profiles that modify significantly the extinction spectra. On the other hand, near-field coupling plays a major role in the broadening of the SLRs. It is reasonable to expect that similarly to a single particle LSPR, the optical losses of an isolated meta-molecule can be compensated by collective effects in the array configuration. However, in the case of the meta-molecules formed by 54 ± 4 nm nanoparticles the polarizability is significantly more complex, and due to the intense near-field exhibited in this case a broad resonance can still be observed (Figure S4, Supporting Information).

This idea can be explored by simulation, comparing the extinction cross-section of different meta-molecules, both isolated and as part of a 2D array (Figure 5A–C), and the corresponding spatial distribution of the electric field of the unit cell of each plasmonic supercrystal (Figure 5D–F).

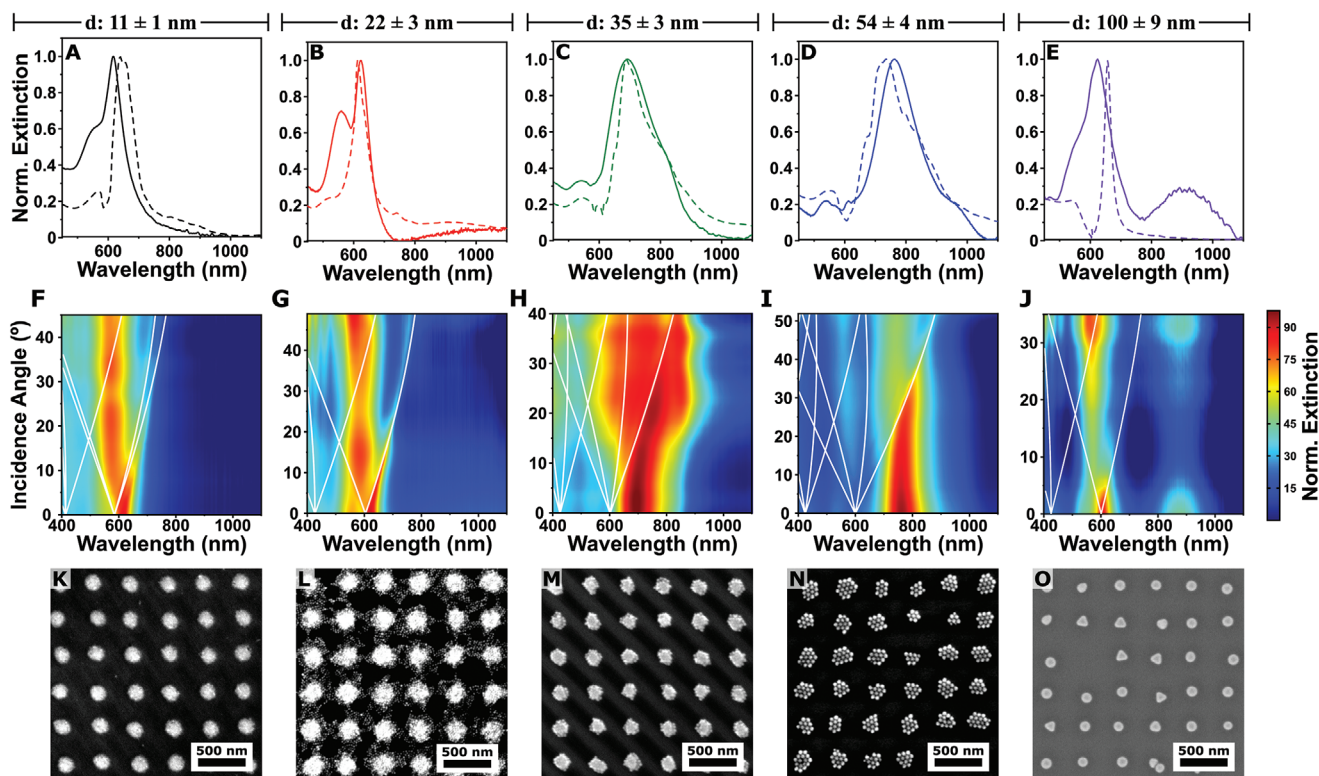


Figure 4. Influence of the diameter of the gold colloid. A–E) Extinction spectra measured at normal illumination incidence, F–J) normalized extinction contour plots as a function of the illumination angle, and K–O) scanning electron microscopy (SEM) images of plasmonic meta-molecule arrays prepared using gold nanoparticle with different diameters: A,F,K) 11 ± 1 nm, B,G,L) 22 ± 3 nm, C,H,M) 35 ± 3 nm, D,I,N) 54 ± 4 nm, and E,J,O) 98 ± 6 nm. The contour plots have been normalized to the extinction maxima. The lattice parameter is 400 nm for all the samples. The white solid lines are analytical calculations of the Wood's anomalies angular dependence. Additional SEM images at different magnifications can be found in Figure S7 in the Supporting Information.

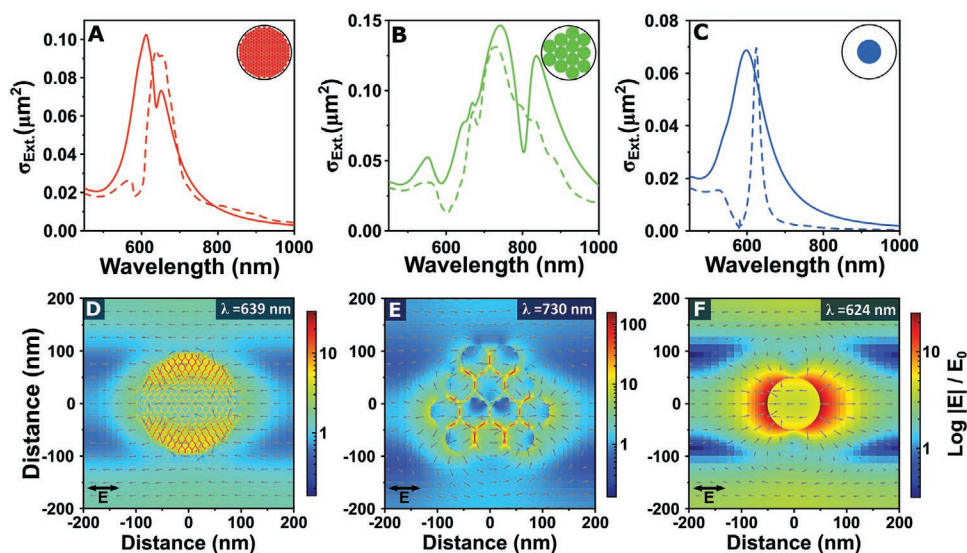


Figure 5. A–C) Simulated extinction spectra of isolated (solid lines) and 2D square arrays with 400 nm pitch (dotted lines) of: meta-molecules composed of A) 11 and B) 54 nm nanoparticles; and C) single 98 nm nanoparticles. Insets: Schematic representation of the simulated geometries; the black line represents the template cavity. D–F) Simulated electric field distribution at the extinction maxima for plasmonic arrays composed of: meta-molecules composed of D) 11 and E) 54 nm nanoparticles; and F) single 98 nm nanoparticles.

The calculated extinction cross-section of the 54 nm meta-molecules resulted to be much higher and broader than the ones relative to 11 or 98 nm particles, confirming the major role played by near-field coupling in the extinction profile of the 2D arrays.

Close inspection of the electric field distribution for the three cases corroborates our hypothesis. In fact, meta-molecules composed of 11 nm nanoparticles (Figure 5D) present a larger number of less intense near-field hot spots, in agreement with the reduced scattering strength observed for smaller gold nanoparticles (Figure S15, Supporting Information). On the contrary, for 54 nm nanoparticles a much higher electric field is concentrated in the hot-spots between neighboring particles (Figure 5E), further increasing optical losses of the relative SLR. In the case of plasmonic lattice of single 98 nm nanoparticles, the values of the electric field and extinction cross-section are similar to the ones obtained for 11 nm nanoparticle case (Figure 5C,F) supporting the idea of creating plasmonic supercrystals out of small colloids with optical properties akin to the large single particle counterparts.

In order to illustrate the tunability in position and spectral bandwidth that can be achieved with the template induced assembly of different colloids in combination with different lattice parameters, we summarized the extinction spectra taken under normal incidence in Figure 6. A clear trend appears, where plasmonic supercrystals prepared with smaller nanoparticles display considerably narrower SLRs. Even though theoretical predictions indicate that single nanoparticle arrays would still sustain the highest quality factors,^[16] 2D plasmonic arrays prepared with small nanoparticles are practically much easier to prepare compared to arrays composed of single nanospheres, which are heavily affected by concentration gradient and normally present a high number of defects (vacancies in the array, misplacement of nanoparticles, aggregates, etc.). Moreover, there is an inevitable distribution in the position of each nanoparticle within its cavity, ultimately translating in lower experimental quality factors. Indeed, if we keep this in consideration when simulating the optical response of single nanoparticles array, purposely introducing vacancies and positional distribution we observed a significant detrimental effect

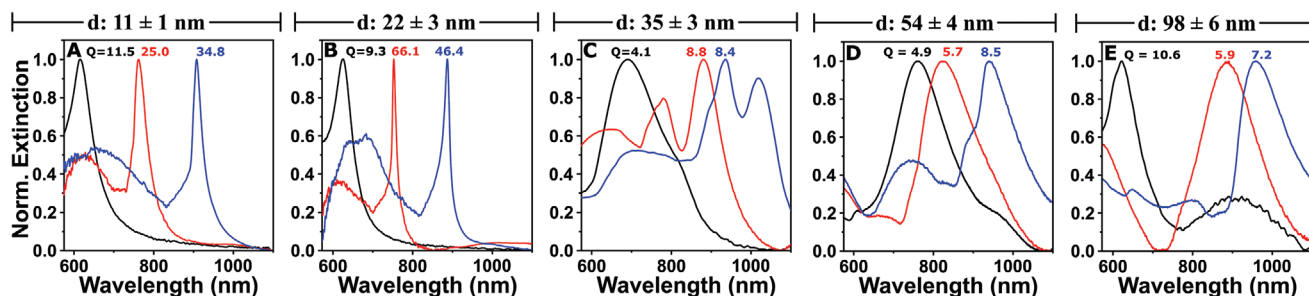


Figure 6. A–E) Extinction spectra collected at normal incidence of plasmonic meta-molecules 2D arrays fabricate using square arrays with lattice parameters of 400 (black solid lines), 500 (red solid lines), and 600 nm (blue solid lines) and gold nanoparticles of different sizes: A) 11 ± 1, B) 22 ± 3, C) 35 ± 3, and D) 54 ± 4, E) 98 ± 6 nm.

on the quality factor of the SLR (Figure S13, Supporting Information). Another interesting point to consider is how the use of smaller colloids allows for the fabrication of more homogenous clusters. In fact, a higher value of the parameter Γ (introduced in the Supporting Information Table S3), achieved either by an increase of the hole diameter (Φ) or a decrease of the nanoparticle diameter (d), translates in an increased homogeneity of the meta-molecules composition. In Figures S11 and S12 in the Supporting Information, we analyze how this effect reduces the impact of geometrical differences between meta-molecules, as the plasmonics of each repeating unit is not drastically modified by defect in its internal structure, such as vacancies or misplacement of single building blocks.

The case of $d = 11$ nm deserves special attention; when PEG chain with a molecular weight of 2000 g mol^{-1} was used for the coating of the colloid, we were able to obtain sharp SLRs, but characterized by a much lower intensity (Figure S14, Supporting Information). This can be explained considering the large interparticle distance (compare to the diameter of the single nanoparticles), resulting in an insufficient scattering generated by each meta-molecule, with a detrimental effect on the far-field coupling (Figure S15, Supporting Information).^[45] For this reason, we utilize a shorter PEG chain (800 g mol^{-1}) for the specific case of 11 nm colloids (Figure 6A). This allowed us to obtain more intense SLRs, even though slightly broader than the ones observed for $d = 22$ nm, probably due to the increased near-field coupling, inducing higher optical losses.

Overall, both near-field and far-field considerations point toward the importance of engineering near-field coupling inside the repeating unit as a key parameter to control the optical properties of complex plasmonic lattice, and how a better control over interparticle separation represents an interesting research direction to be pursued in the future. Our findings allowed us to measure lattice resonances with quality factors as high as $Q = 66$, when using gold colloids of 22 nm in size for our meta-molecule-based plasmonic supercrystals, one of the highest reported in the literature for plasmonic arrays prepared through colloidal self-assembly of gold nanoparticles. Therefore, whereas gold nanoparticles with a diameter above 50 nm are to be preferred when looking for strong near-field coupling, plasmonic arrays prepared with smaller colloids can lead to easily accessible and high-quality factor plasmonic cavities.

3. Conclusions

The lack of control over near-field coupling in template-mediated colloidal self-assembly represents one of the major challenges preventing this approach to compete with top down fabrication methods. In this paper, we show how taking advantage of the versatility of the technique for the implementation of colloids with different sizes is possible to engineer the plasmonic coupling within each meta-molecule to tune the optical response of the prepared photonic plasmonic supercrystals. In particular, by reducing the diameter of the colloidal building block below 25 nm, the optical losses are reduced, opening up the possibility of obtaining narrow resonances using colloidal template-induced assembly technique. Our approach allowed

us to routinely prepare plasmonic arrays capable of producing lattice resonances with quality factors up to 66. These are among the highest reported to date in relation to lattice resonances in the visible that are prepared through colloidal self-assembly.^[31] These narrow features can be further improved studying the interparticle distance (with the aid of longer or shorter ligands, or different core-shell structures), improved index matching schemes, different metal colloids (like silver), or optimized measurement conditions.^[18] Other interesting direction would be to modify the nanoparticle ligand shell after the preparation of the colloidal lattices,^[46] or modify the internal packing of the building blocks changing their shape during synthesis.^[47] Ultimately, the choice of building block dimensions dictates the optical properties of the plasmonic supercrystals, and will also modify the contributions of scattering and absorption components to the extinction spectra (Figure S16, Supporting Information). These aspects have important implications in the preparation of plasmonic arrays for specific applications. On one side, bigger colloids produce plasmonic crystals with strong near-field coupling and a broadband absorption, perfectly suitable for applications in super absorption, catalysis, or plasmonic heating. On the other side, small-colloid-based plasmonic supercrystals lead to narrower optical features that can be used to prepare tunable plasmonic cavities, targeting lasing or sensing, for instance.^[48] In sum, we find these results encouraging for the future of the template-induced assembly technique, and we believe there is still much to be explored toward the implementation of this cheap and scalable approach for the fabrication of plasmonic devices.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

colloids, plasmonics, soft lithography, template induced self-assembly

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